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Part 2: Logic, Ontology, Expert System, and Knowledge Graph

SwiftRoute Delivery Risk Assessment

2.1 Propositional Logic

Propositional Formulas

Let the atomic propositions be:

- W: Weather score exceeds 0.7
- M: The delivery is a motorcycle delivery
- H: The delivery is in a high-density zone
- D: The delivery distance exceeds 50 km
- F: The delivery is flagged for manual review
- S: A supervisor is notified
- V: The vehicle is a truck

Statements in propositional logic:

1. A delivery is flagged if the weather score exceeds 0.7 and it is a motorcycle delivery.

$((W \text{ AND } M) \rightarrow F)$

2. A delivery is flagged if it is in a high-density zone or the distance exceeds 50 km.

$((H \text{ OR } D) \rightarrow F)$

3. If a delivery is flagged, a supervisor is notified, unless the vehicle is a truck.

$((F \text{ AND NOT } V) \rightarrow S)$

9.Truth Table for First Rule Rule: $((W \text{ AND } M) \rightarrow F)$

W	M	W AND M	F	(W AND M) $\rightarrow$ F
T	T	T	T	T
T	T	T	F	F
T	F	F	T	T
T	F	F	F	T
F	T	F	T	T
F	T	F	F	T
F	F	F	T	T
F	F	F	F	T

The rule is false only when both W and M are true but F is false.

10. Inference Using Resolution

Premises:

4. $W = \text{True}$
5. $M = \text{True}$
6. $((W \text{ AND } M) \rightarrow F)$

$((W \text{ AND } M) \rightarrow F)$ is equivalent to $\text{NOT}(W \text{ AND } M) \text{ OR } F$.

Using De Morgan's law:

$\text{NOT } W \text{ OR } \text{NOT } M \text{ OR } F$

Now use the facts:

- Since W is true, $\text{NOT } W$ is false.
- Since M is true, $\text{NOT } M$ is false.

The clause becomes:

$\text{False OR False OR } F$

Therefore:

$F = \text{True}$

So, the delivery is flagged.

2.2 First-Order Logic

Predicates and Constants

Predicates:

- $\text{HighDensityZone}(x)$: delivery x is assigned to a high-density zone
- $\text{HasGPS}(x)$: vehicle or driver x has a GPS tracker
- $\text{Shortest}(r)$: route r is the shortest route
- $\text{MostCongested}(r)$: route r is the most congested route
- $\text{Driver}(d)$: d is a driver
- $\text{Assigned}(d, o1, o2)$: driver d is assigned to deliveries $o1$ and $o2$ at the same time

Constants:

- Van07 : a specific vehicle
- RouteA : a specific route
- Driver1 : a specific driver
- $\text{Order12}, \text{Order18}$: example deliveries

Statements in First-Order Logic

- Every vehicle assigned to a high-density zone must carry a GPS tracker.

forall x (HighDensityZone(x) -> HasGPS(x))

- There exists a route that is both the shortest and the most congested.

exists r (Shortest(r) AND MostCongested(r))

- No delivery driver can be assigned to two simultaneous deliveries.

forall d forall o1 forall o2 ((Driver(d) AND Assigned(d, o1, o2)) -> o1 = o2)

Modus Ponens and Universal Instantiation

Given:

- forall x (HighDensityZone(x) -> HasGPS(x))
- HasGPS(Van07) = True
- HighDensityZone(Van07) = True

Using Universal Instantiation on premise 1 with x = Van07:

HighDensityZone(Van07) -> HasGPS(Van07)

Given:

- HighDensityZone(Van07) = True
- Apply Modus Ponens:
- HasGPS(Van07) = True
- New fact derived:
- Van07 has a GPS tracker.

11. Why First-Order Logic Is More Expressive

First-order logic is more expressive than propositional logic because it can represent objects, properties of objects, and relationships between objects. Propositional logic treats each statement as a whole true-or-false unit, such as W for bad weather or F for flagged delivery. It cannot clearly express rules about all vehicles, some routes, or relationships between drivers and deliveries. First-order logic can use variables, predicates, and quantifiers such as forall and exists. For example, the rule forall x (HighDensityZone(x) -> HasGPS(x)) states that every vehicle or delivery assigned to a high-density zone must have GPS. This directly supports the SwiftRoute examples because delivery operations involve many vehicles, routes, zones, drivers, and orders. First-order logic is therefore better for general rules that apply across many real-world objects.

2.3 Ontology Design

12. Classes, Object Properties, and Data Properties

Classes:

- i. Vehicle
- ii. Truck
- iii. Motorcycle
- iv. Route
- v. Delivery
- vi. Personnel
- vii. Driver
- viii. Supervisor
- ix. Zone
- x. Customer

Object properties:

- xi. assignedToDriver(Delivery, Driver)
- xii. usesVehicle(Delivery, Vehicle)
- xiii. travelsOnRoute(Delivery, Route)
- xiv. locatedInZone(Delivery, Zone)
- xv. supervisedBy(Driver, Supervisor)

Data properties:

- xvi. hasDistanceKm(Route, decimal)
- xvii. hasWeatherScore(Delivery, decimal)
- xviii. hasDeliveryStatus(Delivery, string)
- xix. hasGPSStatus(Vehicle, boolean)

13. Class Hierarchy Diagram

Class Hierarchy for SwiftRoute Ontology

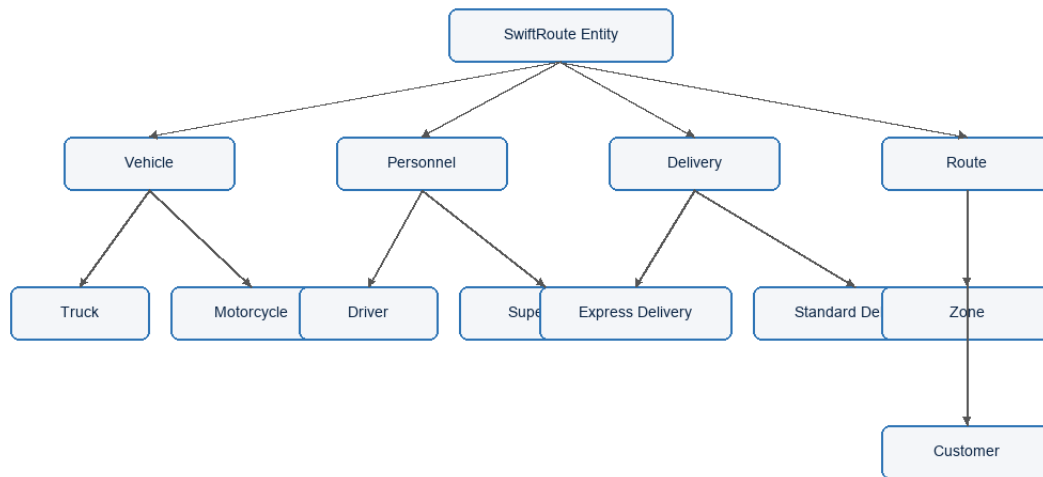


Figure 1. SwiftRoute ontology class hierarchy.

Minimum required hierarchy:

- Vehicle has subclasses Truck and Motorcycle.
- Delivery has subclasses ExpressDelivery and StandardDelivery.
- Personnel has subclasses Driver and Supervisor.

14. SPARQL-Like Queries

Query 1: Find deliveries in high-density zones without GPS-enabled vehicles.

SELECT	?delivery	?vehicle	?zone
WHERE			{
?delivery	locatedInZone	?zone	.
?zone	zoneType	"HighDensity"	.
?delivery	usesVehicle	?vehicle	.
?vehicle	hasGPSStatus	false	.
}			

This query helps managers identify risky deliveries that violate the GPS requirement for high-density zones.

Query 2: Find long-distance deliveries that were flagged for review.

```
SELECT      ?delivery      ?route      ?distance
WHERE
  ?delivery      travelsOnRoute      ?route      .
  ?route      hasDistanceKm      ?distance      .
  ?delivery      hasReviewStatus      "Flagged"      .
FILTER      (?distance      >      50)
}
```

This query helps managers audit deliveries where distance may have contributed to higher operational risk.

Query 3: Find drivers supervised by a specific supervisor and their assigned deliveries.

```
SELECT      ?driver      ?delivery
WHERE
  ?driver      supervisedBy      SupervisorA      .
  ?delivery      assignedToDriver      ?driver      .
}
```

This query helps a supervisor monitor all deliveries assigned to their drivers.

2.4 Expert System and Knowledge Graph

14. Rule-Based Expert System for Delivery Risk Assessment

Rules in plain English:

- If the weather score is greater than 0.7 and the delivery uses a motorcycle, then classify the delivery as high risk and flag it for manual review.
- If the delivery is in a high-density zone and the vehicle does not have GPS, then classify the delivery as high risk and flag it for manual review.
- If the route distance is greater than 50 km, then classify the delivery as medium risk.
- If the delivery is flagged and the vehicle is not a truck, then notify a supervisor.
- If the customer rating is below 3 and the delivery has more than 4 customer care calls, then classify the delivery as medium risk.
- If the product importance is high and the delivery is already medium risk, upgrade it to high risk.
- If the delivery is high risk, assign it to manual review.
- If the delivery is low risk, continue normal dispatch.

Example delivery:

- Delivery ID: D102
- Weather score: 0.82
- Vehicle: Motorcycle

- Zone: HighDensity
- GPS: True
- Route distance: 32 km
- Product importance: Medium

Reasoning trace:

- Weather score is 0.82, which is greater than 0.7.
- Vehicle is Motorcycle.
- Rule 1 fires: weather is risky and the delivery uses a motorcycle.
- Delivery D102 is classified as High Risk.
- Delivery D102 is flagged for manual review.
- Since the vehicle is not a truck, Rule 4 fires.
- A supervisor is notified.

Final classification: High Risk - Manual Review Required.

Limitations of an expert system compared with an ML classifier from Part 1:

- Expert systems depend on manually written rules, so they may miss hidden patterns in large datasets.
- They can become difficult to maintain when many rules interact or conflict.
- They do not automatically improve from new delivery records unless humans update the rules.
- Thresholds such as weather score 0.7 or distance 50 km may be too rigid.
- They are easier to explain than many ML models, but usually less adaptive.

15. Knowledge Graph

Knowledge Graph: Delivery D102

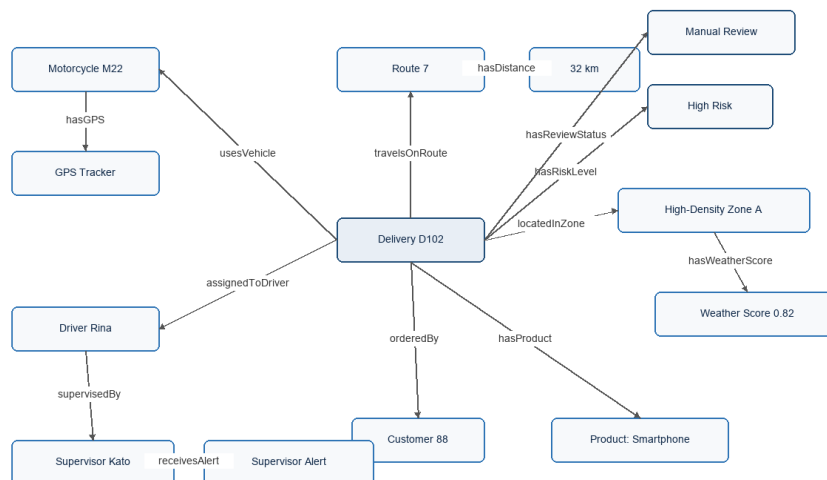


Figure 2. Knowledge graph for Delivery D102.

Nodes included:

- i. Delivery D102
- ii. Motorcycle M22
- iii. Driver Rina
- iv. Route 7
- v. High-Density Zone A
- vi. Customer 88
- vii. Product: Smartphone
- viii. GPS Tracker
- ix. Supervisor Kato
- x. 32 km
- xi. Weather Score 0.82
- xii. High Risk
- xiii. Manual Review
- xiv. Supervisor Alert

Example insights from graph traversal:

Starting from Delivery D102, a manager can traverse to Zone A, then to Weather Score 0.82, and then to High Risk. This reveals that weather and zone conditions contributed to the risk classification.

Starting from Supervisor Kato, the manager can traverse to Driver Rina, then to Delivery D102, and then to Manual Review. This shows which supervisor is responsible for responding to the flagged delivery.